$ ./system_boot --profile=codernex --v=2.0

CODERNEX (BORHAN UDDIN)

Full-Stack Software Engineering, Scalable Backend Systems & AI Automation

LOCATION
Khulna, BD (Remote)

URL
https://codernex.dev

EMAIL
borhan.dev@gmail.com

WHATSAPP
(+88) 01888527959

1. Executive Summary & Founder Profile

Specialized Full-Stack & Systems Architecture Practice

Codernex is the specialized software engineering and system architecture brand operated by Borhan Uddin, Software Engineer II and systems architect. Codernex bridges architectural vision with execution-grade code, delivering high-throughput backend services, autonomous AI agentic workflows, and modern web applications.

Borhan Uddin currently serves as Software Engineer II at EWWFL (Tampa, FL - Remote), leading payment system migrations (Stripe), high-throughput data sync engines (SFTP, Gmail API, AWS S3), and full-scale frontend SSR caching achieving a 140% improvement in Google Lighthouse performance scores (from 45 to 95+). Previously, he led engineering teams at Softronixs System LTD and Borno IT, developing multi-vendor e-commerce backends maintaining 99.9% uptime.

Core Technical Competencies

- **Backend & Database Engineering:**
Redis, MongoDB, SQLAlchemy, TypeORM, Prisma, AsyncPG, BullMQ, Celery.
- **AI Agents & Intelligent Workflows:**
(Retrieval-Augmented Generation), Vector Similarity Search, Autonomous Tool Calling.
- **Frontend & User Experience:**
Tailwind CSS, RTK Query, TanStack Query, ReactFlow visual DAG canvases.
- **DevOps, Cloud & Infrastructure:**
kernel optimization, CI/CD automated build pipelines, Nginx.

2. Comprehensive Professional Services Catalog

End-to-End Engineering, System Architecture & Technical Consulting

Codernex provides specialized engineering services tailored for high-growth startups, scale-ups, and established enterprises requiring bulletproof architecture, zero tech debt, and rapid execution.

Service A: Scalable Backend Architecture & Microservices

ENTERPRISE BACKEND

- **Modular Backend Design:** adhering to SOLID principles and Clean Architecture.
- **Database & Vector Storage:** connection pooling, and pgvector semantic embeddings search.
- **High-Throughput Pipelines:** worker thread concurrency.
- **Security & Authentication:**

Service B: Autonomous AI Agents & Intelligent Workflow Engines

AI & AUTOMATION

- **Agentic DAG Orchestration:** configuration of autonomous agents.
- **Enterprise RAG Systems:** LLM providers for private document search.
- **Tool Calling & Sandboxes:** real-time SSE execution telemetry streaming.
- **Multi-Agent Collaboration:**

Service C: Modern Frontend Engineering & Interactive Web Apps

MODERN FRONTEND

- **Next.js & React 19 Mastery:** Actions, and incremental static regeneration.
- **Complex Interactive UIs:** custom rich-text block editors.
- **Optimized Client State:** Query.
- **Modern Responsive Design:**

Service D: Performance Engineering, Audits & Optimization

PERFORMANCE LAB

- **Core Web Vitals & Lighthouse:** achieve 95+ Lighthouse scores.
- **CPU-Bound Bottleneck Removal:** main-thread locking.
- **Query & Index Optimization:** multi-tiered Redis caching.
- **Media & Asset Compression:**

Service E: E-Commerce, Payment Systems & Third-Party Integration

PAYMENTS & APIS

- **Payment Infrastructure:** fraud-resistant validation.
- **Third-Party API Sync:** cloud storage (S3 / R2).
- **Invoicing & Billing Engines:** recurring payment handlers.

3. Featured Architectural Case Studies

Demonstrated Track Record of Scalable System Implementations

Case Study 1: Agentic Workflow Automation Engine

CORE PRODUCT

- **Architecture:**
execution sandboxes, and DAG workflows.
- **Tech Stack:**
Face smolagents, Mailtrap SDK, Turborepo.
- **Key Features:**
verification, and GDPR-compliant audit logs.
- **Live Showcase:**

Case Study 2: Codernex PostScheduler

AI SCHEDULER

- **Architecture:**
prompt scheduling.
- **Tech Stack:**
Background Queue Workers.
- **Key Features:**
human-in-the-loop review, and automated publishing.
- **Repository:**

Case Study 3: Freelancer Invoice Management System

FINTECH SAAS

- **Architecture:**
agencies.
- **Tech Stack:**
- **Key Features:**
metrics analytics dashboard.
- **Live Demo:**

Case Study 4: NexPress CLI - Parallel Image Compression

OPEN SOURCE

- **Architecture:**
web servers.
- **Tech Stack:**
- **Key Metrics:**
event loop.

Case Study 5: KAGOZ - Business Management & Analytics SaaS

ENTERPRISE SAAS

- **Architecture:**
anonymous visitor engagement tracking.
- **Tech Stack:**

4. Engagement Lifecycle & Quality Commitments

Engineering Standards, Milestones & Operational SLA

Standard 5-Phase Project Delivery Framework

- **Phase 1: Architecture Discovery & Requirement Specification:** technology stack finalization, and Statement of Work (SOW) agreement.
- **Phase 2: Database Schema & Core Architecture Blueprint:** indices, authentication models, and system sequence workflows.
- **Phase 3: Sprint Execution & Continuous Integration:** unit/integration testing, and staging environment deployments.
- **Phase 4: Security Audit, Performance Profiling & QA:** Lighthouse performance optimization (target 90+), and stress testing.
- **Phase 5: Production Deployment, Monitoring & Handover:** configuration, Cloudflare protection, complete API documentation, and handover walkthrough.

Service Level Commitments (SLA)

- **Uptime & Reliability Target:** availability with automated health checks.
- **Post-Delivery Bug-Fix Warranty:** delivered source code to resolve any deviations from agreed specs.
- **Response Time SLA:** responses within 4 hours; standard queries within 24 hours.
- **Code Quality & Maintainability:** 100% type safety, and zero lint warnings.

5. Terms and Conditions

Official Commercial Terms & Service Agreement

These Terms and Conditions govern all software development, technical consulting, architecture design, and maintenance services provided by Codernex (Borhan Uddin) to the client ('Client'). By initiating a project, executing a Statement of Work (SOW), or submitting payment, the Client agrees to be bound by these provisions.

- **Article 1: Scope of Work & Deliverables:**
Statement of Work, quotation, or technical proposal. Any modifications, additional features, or out-of-scope requests will be quoted separately through a formal written Change Order.
- **Article 2: Intellectual Property & Ownership:**
intellectual property rights, copyright, and title to custom source code, documentation, and assets developed specifically for the Client shall transfer entirely to the Client. Codernex retains ownership of pre-existing proprietary libraries, developer toolkits, and open-source frameworks utilized, granting the Client a perpetual, royalty-free, worldwide license to utilize them within the delivered software.
- **Article 3: Invoicing, Payments & Taxes:**
otherwise specified in the SOW. Payments may be remitted via Stripe, direct bank wire transfer, or approved digital settlement methods in USD or agreed local currency. Delinquent balances exceeding thirty (30) days may result in temporary suspension of active services.
- **Article 4: Confidentiality & Non-Disclosure (NDA):**
technical architectures, database records, customer data, and trade secrets shared during the engagement as strictly confidential. Neither party shall disclose Confidential Information to third parties without prior written consent, surviving project completion for a period of three (3) years.
- **Article 5: Warranties & Limitation of Liability:**
substantially in accordance with agreed functional specifications for thirty (30) calendar days following handover. Except for deliberate misconduct, Codernex's aggregate liability under any circumstance shall not exceed the total fees paid by Client for the specific engagement.
- **Article 6: Termination & Offboarding:**
days written notice. In the event of early termination, the Client shall pay for all work satisfactorily completed up to the termination date, and Codernex shall provide all completed source code and assets.
- **Article 7: Governing Law & Dispute Resolution:**
applicable commercial laws. In the event of any disagreement, the parties agree to seek amicable resolution through good-faith mediation before pursuing formal legal proceedings.

6. Comprehensive Privacy Policy

Data Protection, Security Measures & Regulatory Compliance

Codernex ('we', 'us', or 'our') is committed to upholding the highest privacy, data protection, and confidentiality standards. This policy outlines how we handle personal data, technical credentials, and proprietary code across all our services and interactions.

- **1. Information We Collect:**
deliver software projects. This includes: (a) Contact data (name, email address, WhatsApp/phone number), (b) Business requirements and technical specifications, and (c) Secure development credentials (API keys, staging database URLs, server access keys) provided by the client for deployment.
- **2. Purpose & Legal Basis of Processing:**
engineering and architectural services, (b) Executing secure automated build and deployment pipelines, (c) Responding to inquiries and technical support requests, and (d) Fulfilling accounting, invoicing, and legal obligations.
- **3. Secure Credential Handling & Data Security:**
secrets, API tokens, and access credentials must be transmitted via encrypted channels (Vault, 1Password, or encrypted environment files). Credentials are never committed to public repositories, and all staging environments utilize AES-256 encryption at rest.
- **4. Third-Party Service Providers:**
delivery of our services, including: AWS (cloud storage), Cloudflare (DNS and R2 object storage), Stripe (payment processing), and GitHub (private repository hosting). Each third party complies with international security frameworks including SOC2 and GDPR.
- **5. Data Retention & Erasure (GDPR Compliance):**
securely deleted thirty (30) days following project handover and warranty conclusion. Clients have the right at any time to request complete erasure of their stored data by contacting us directly.
- **6. Data Protection Inquiries:**
contact Borhan Uddin directly at borhan.dev@gmail.com.

7. Official Contact & Connection Channels

Direct Communication for Projects, Architecture & Advisory

Get In Touch for Your Next Project

VERIFIED TRANSMISSION

- **Founder & Principal Engineer:**
- **Official Website:**
- **Direct Email Address:**
- **WhatsApp / Mobile:**
- **Physical Base Location:**
- **GitHub Organization:**
- **Engineering Articles & Blog:**
- **Interactive Project Showcase:**